K. J. Somaiya School of Engineering

EXCP · CAPSTONE MINI PROJECT

Stroke Risk Factor Analysis Using Patient Health Records

Python Programming Laboratory · Statistical analysis

Student / course	Details
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Abstract

A public stroke benchmark containing 5,110 records and 12 variables was analyzed using Python. The primary analysis uses 4,254 adults, including 247 records with stroke. Age, average glucose and hypertension show associations with the recorded outcome. Adult mean BMI differs by only 0.04 kg/m² (95% CI −0.86 to 0.94). Smoking status shows a small association that is sensitive to the records included. Missing BMI, unknown smoking status, unequal outcome groups and limited source provenance constrain interpretation. This project demonstrates reproducible statistical analysis; it does not establish causes or provide a clinical prediction tool.

Prepared: 25 September 2026

1. Problem definition and objectives

Problem statement

Analyze the statistical associations between recorded stroke status and age, average glucose level, BMI, hypertension and smoking status using a public patient health dataset. Clean and validate the data, summarize the variables, visualize group differences, and apply suitable statistical tests with effect sizes and uncertainty. Assess how missing values and age composition affect the findings, without claiming causation or predicting an individual patient's future stroke risk.

Why this fits EXCP

The project applies programming, numerical computation and data visualization to the healthcare domain. It turns a tabular health dataset into an auditable analysis through modular functions, validation checks, statistical testing and a report.

Objectives

- Select a public dataset with all five requested factors and an outcome label.
- Identify missing values, duplicate IDs, unexpected categories and unusual measurements.
- Use NumPy and Pandas to create summaries and meaningful analysis populations.
- Create charts with appropriate denominators and uncertainty.
- Compare stroke and non-stroke records using five primary statistical tests.
- Check whether conclusions change with age restrictions and incomplete records.
- Deliver runnable source code, a notebook, a report and viva preparation.

Research scope

"Risk factor" describes candidate characteristics associated with the recorded stroke label. The dataset has no follow-up dates or measurement timing, so it cannot establish temporal order, causation, incidence, or future risk. All comparisons are unadjusted for other covariates. A forecasting task would require a time variable, which this dataset does not provide.

2. Dataset and data dictionary

Stroke Prediction Dataset [1, 2]

The supplied raw CSV has 5,110 rows, 12 columns, 249 records labeled stroke and 4,861 labeled no stroke. It contains all five requested explanatory variables. The original landing page is Kaggle; the exact analyzed CSV was acquired from a public GitHub mirror. The file is approximately 310 KiB and meets the form's 10 MB limit.

Column	Meaning	Analysis role
id	Record identifier; not a health measurement	Traceability only
gender	Female, Male or Other	Retained; no primary test
age	Age in years; fractions occur for children	Primary continuous factor
hypertension	0 absent; 1 present	Primary categorical factor
heart_disease	0 absent; 1 present	Supplementary description
ever_married	Yes / No	Retained
work_type	Five recorded employment categories	Retained
Residence_type	Urban / Rural	Retained
avg_glucose_level	Recorded average glucose; unit unverified	Primary continuous factor
bmi	Body mass index; conventional kg/m^2	Primary continuous factor
smoking_status	Never, former, current, Unknown	Primary categorical factor
stroke	0 no recorded stroke; 1 recorded stroke	Outcome label

Provenance limits

Hospital, geography, collection dates, sampling method, clinical adjudication and the exact glucose measurement protocol were not independently verified. Treat the data as an educational patient-record benchmark, not a representative population survey. No open-license claim is made; consult the source terms before redistribution outside the course. Do not infer patient identity from record IDs.

3. Preparation and analysis population

Cleaning rules

Read N/A as missing BMI. Validate numeric fields, permitted categories, binary labels and positive observed glucose/BMI. Check exact duplicates and conflicting repeated IDs. Preserve unusual but valid observations and flag adult 1.5×IQR values. Keep “Unknown” smoking distinct from “never smoked”. No record is relabeled and no outcome balancing or oversampling is performed.

Stage / data issue	Records
Raw records	5110
Exact duplicates removed	0
Age under 18: retained in raw data; excluded from primary analysis	856
Adult primary population	4254
Adults with recorded stroke	247
Adult BMI missing	181
Adult smoking unknown	862
Adults with observed BMI and known smoking	3256

Why adults are the primary population

The dataset mixes children and adults. Smoking histories and BMI interpretation differ across these groups. The main analysis is restricted to age ≥18; all ages are preserved and analyzed separately as a sensitivity check. The observed age range does not establish that the sample represents every adult age group.

Observed versus imputed exports

Primary tests use observed values, so BMI has fewer records than age and glucose. Smoking tests exclude the Unknown category but descriptive charts show it. An optional teaching export fills missing adult BMI with the adult median (29.2) and retains a flag; it is never used for reported tests. Median filling can distort variance and is not a solution to missing-data bias.

4. Outcome balance and missing data

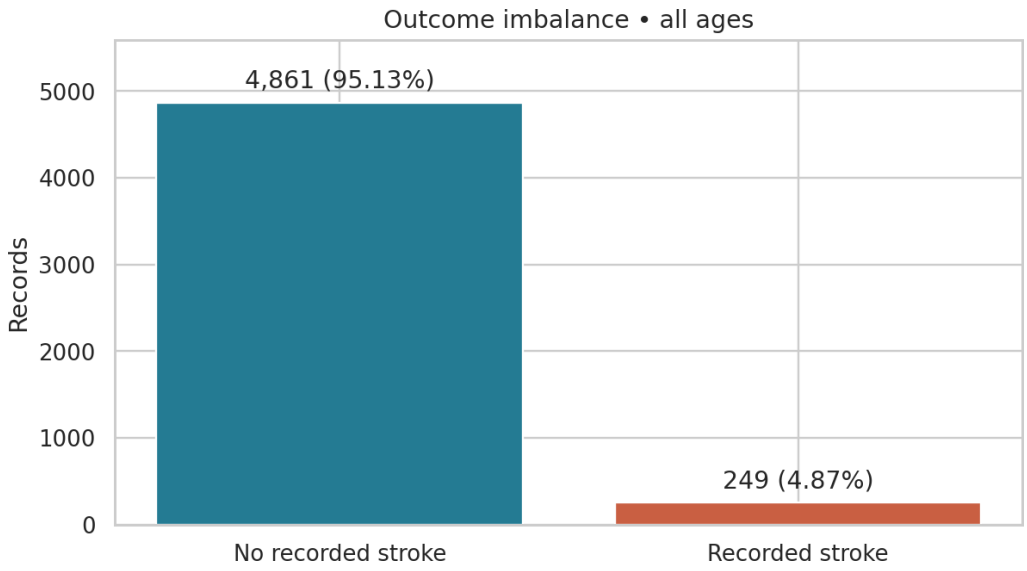


Figure 1. All-age stroke proportion: 249/5,110 = 4.87%. This is a dataset proportion, not a population incidence estimate.

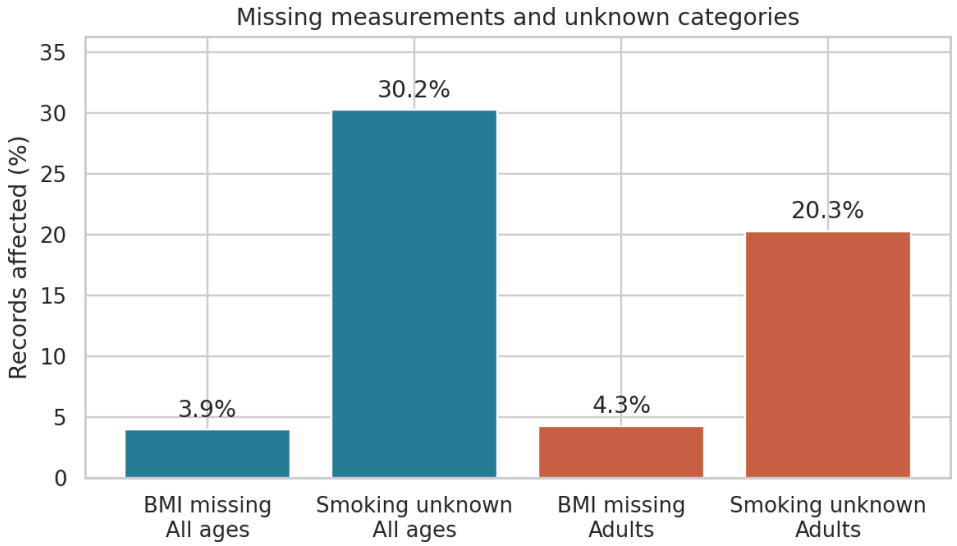


Figure 2. BMI is missing in 201 all-age records; smoking is Unknown in 1,544. In adults, the counts are 181 and 862. Unknown is a recorded category, not an empty CSV cell.

Missingness differs across outcomes

Among adults, BMI is missing in 39/247 stroke records (15.79%) and 142/4,007 non-stroke records (3.54%). This unequal availability may bias observed-BMI comparisons. The missingness mechanism cannot be established from this table. Complete-case analysis is a sensitivity check, not proof that this bias is removed.

5. Statistical analysis plan

Question	Method and reported quantity
Do mean age, glucose and BMI differ?	Two-sided Welch t-tests; mean differences, pointwise 95% CIs and Hedges g.
Is hypertension associated with the label?	Pearson chi-square test without continuity correction; Cramér V.
Does the label vary by known smoking category?	Pearson chi-square across never/former/current smokers; Cramér V. Unknown excluded.
How are five primary tests combined?	Holm correction across exactly five p-values; adjusted $p < 0.05$ marks statistical significance.
How uncertain are descriptive proportions?	95% Wilson intervals for stroke proportions; pointwise, not simultaneous.
Are patterns sensitive to analytic choices?	Mann-Whitney checks, all-age tests and complete adult-record tests.

Hypotheses and assumptions

For each continuous factor, H0 is equality of outcome-group means. For each categorical factor, H0 is independence from the stroke label. Welch tests allow unequal variances and sample sizes. Records are assumed independent, but the source cannot verify independence or random sampling. Welch mean comparisons are supplemented by rank tests because glucose is skewed.

The minimum expected counts are 28.86 for hypertension and 46.45 for smoking, both above 5. The two chi-square tables therefore meet the planned expected-count check. Smoking has 3,392 adults with known status.

Interpretation rules

A small p-value is evidence against the specified null within the analysis assumptions; it is not the probability that the null is true. Statistical significance does not imply a large or causal effect. Confidence intervals and effect sizes are unadjusted and descriptive. The project does not fit a multivariable model, so no factor is described as an independent predictor.

6. Continuous factors: adult comparisons

Factor	Stroke mean (n)	No-stroke mean (n)	Difference; 95% CI
Age	68.21 (247)	49.09 (4007)	19.12; [17.58, 20.67]
Average glucose	133.10 (247)	107.00 (4007)	26.10; [18.22, 33.98]
BMI	30.47 (208)	30.43 (3865)	0.04; [-0.86, 0.94]

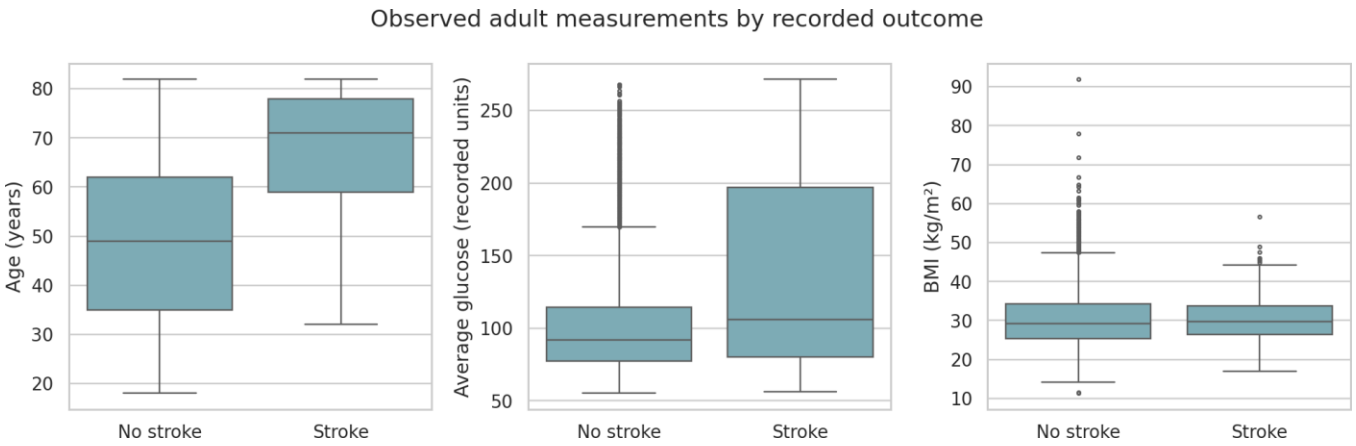


Figure 3. Outliers are displayed and retained. BMI uses observed measurements only.

Factor	Hedges g	Holm-adjusted p	Interpretation
Age	1.108	3.63e-74	Mean difference detected
Average glucose	0.551	1.08e-09	Mean difference detected
BMI	0.005	0.9328	No mean difference detected

Stroke records are approximately 19.12 years older on average and have average glucose 26.10 recorded units higher. BMI shows a very small observed mean difference with an interval spanning zero. This does not establish that BMI has no relationship with stroke; missingness, nonlinearity and other covariates remain relevant limitations.

7. Age and recorded conditions

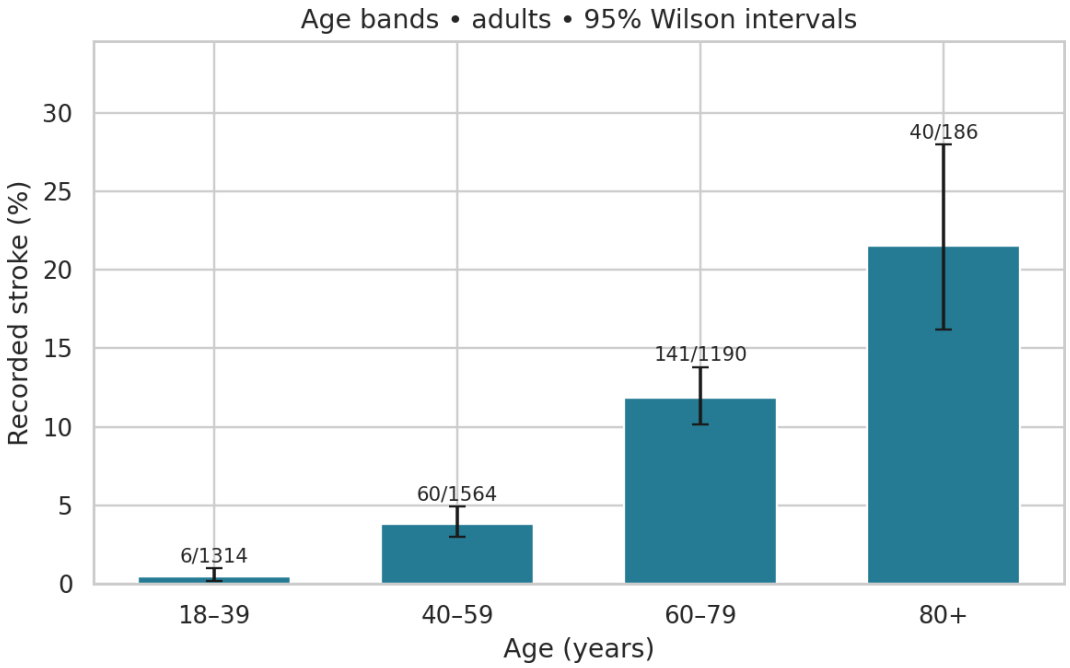


Figure 4. Bars show stroke counts / total records in each adult age band; intervals are 95% Wilson intervals.

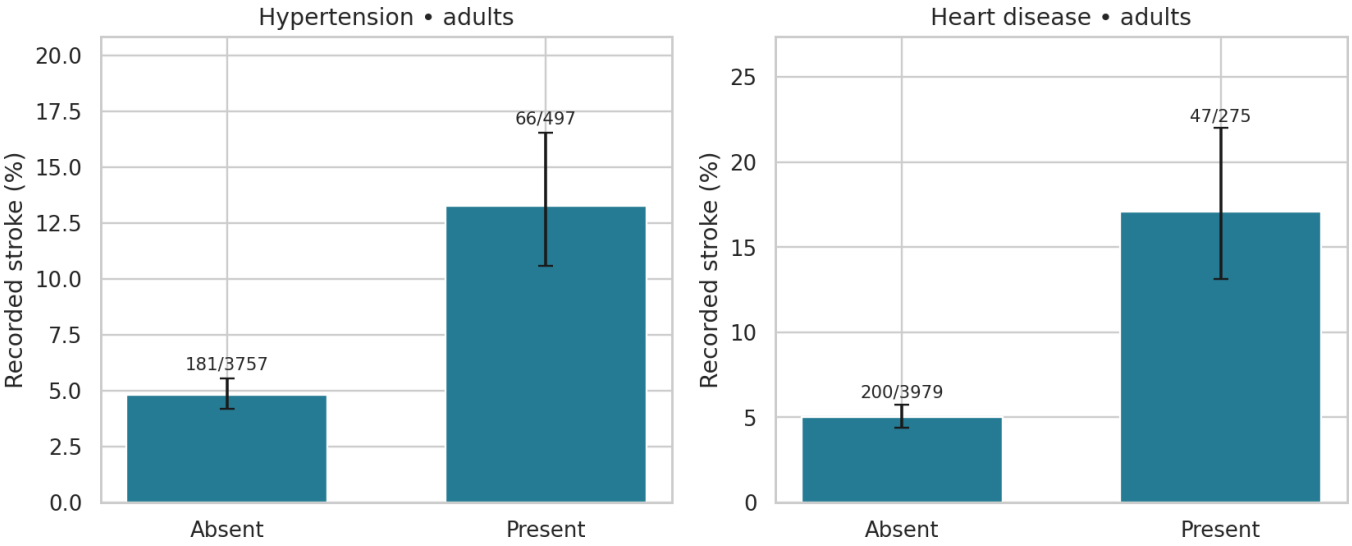


Figure 5. Hypertension is a primary factor. Heart disease is supplementary and has no primary hypothesis test in this project.

Recorded stroke proportions are 13.28% with hypertension and 4.82% without. The hypertension association has Holm-adjusted p = 1.37×10□¹³ and Cramér V = 0.116. These are unadjusted comparisons; differences in age or other factors could account for part of the association. The 80+ band contains 186 records, so its estimate is less precise than larger age groups.

8. Smoking status and interpretation

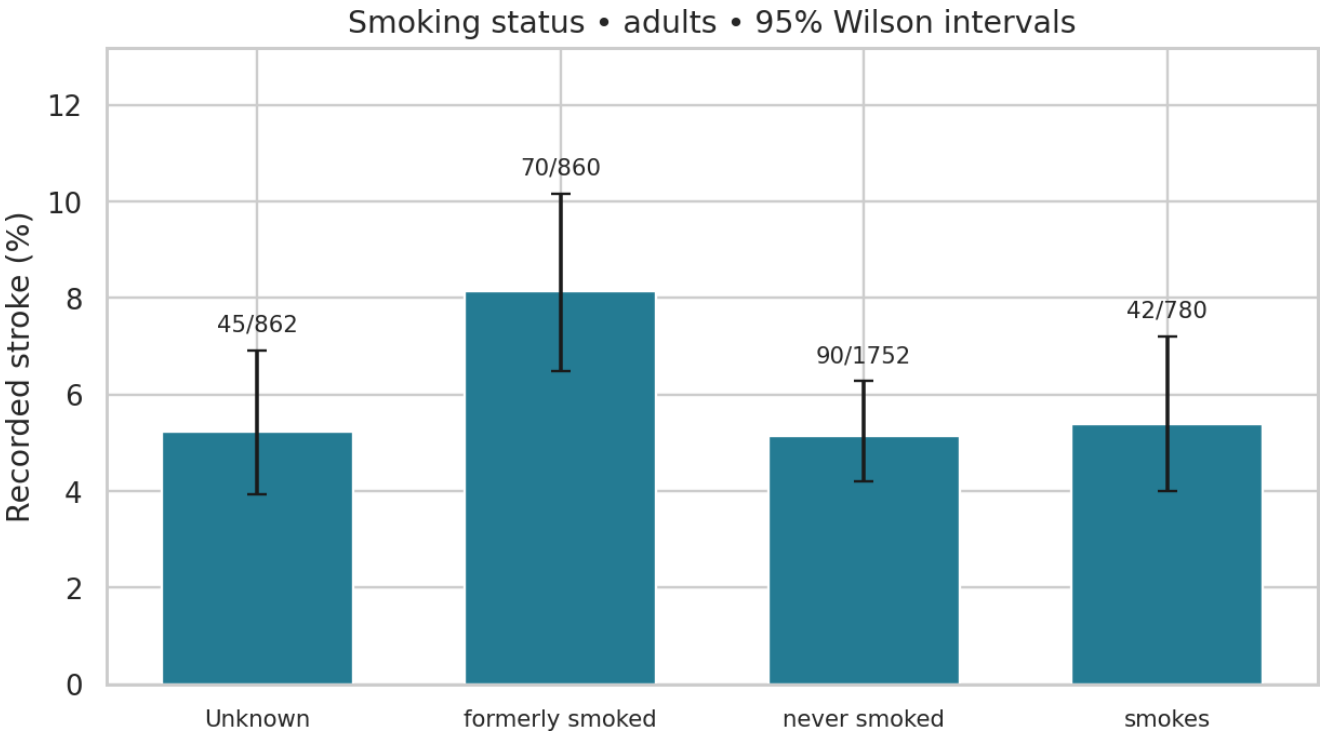


Figure 6. Unknown status is shown for transparency but omitted from the primary smoking test.

Adult category	Stroke / total	Recorded stroke %	Median age
Formerly smoked	70 / 860	8.14	58
Never smoked	90 / 1,752	5.14	49
Smokes	42 / 780	5.38	47
Unknown	45 / 862	5.22	48

Small association, limited robustness

The known-status smoking test has Holm-adjusted p = 0.0143 and Cramér V = 0.054. Former smokers are older on average than current or never smokers; the groups cannot be compared as if all other characteristics were equal. When analysis is restricted to adults with observed BMI and known smoking, adjusted p becomes 0.189. The smoking conclusion is therefore sensitive to which records are included.

These results do not show that current smoking is protective or that quitting increases risk. Smoking history duration, dose, timing relative to stroke and other relevant covariates are unavailable. The exported age-band table helps inspect age composition but is not a multivariable adjustment.

9. Correlations and sensitivity checks

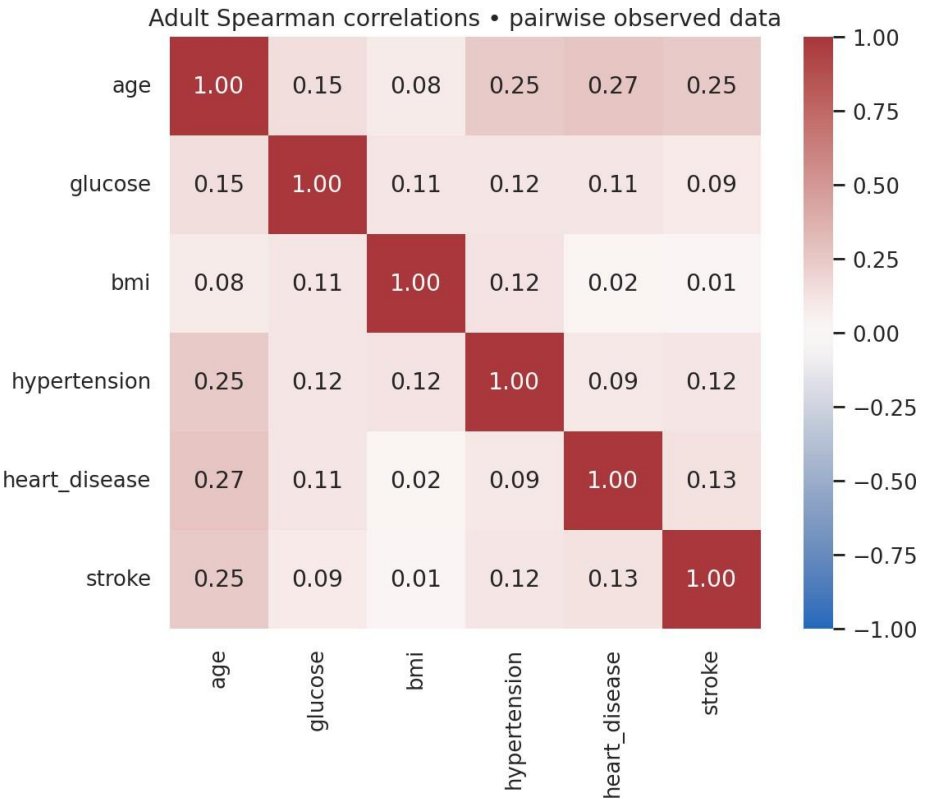


Figure 7. Spearman correlation uses available pairs. Record IDs and arbitrary encodings of nominal categories are excluded. Correlation summarizes pairwise association; it is not causation.

Factor	Primary adult Holm p	All ages Holm p	Complete adults Holm p
Age	3.63e-74	1.06e-94	9.55e-51
Average glucose	1.08e-09	7.20e-11	1.14e-08
BMI	0.9328	6.68e-04	0.7977
Hypertension	1.37e-13	2.43e-19	5.93e-15
Smoking status	0.0143	0.0033	0.1888

BMI appears significant in the all-age analysis but not in the adult analysis, illustrating sensitivity to age composition. Smoking loses significance in complete adult records. Age, glucose and hypertension remain significant in these checks. This is exploratory robustness evidence, not proof that confounding or missingness has been resolved. Rank-test p-values are exported as unadjusted supplementary checks.

10. Implementation and reproducibility

File / folder	Purpose
main.py	Runs preparation, statistics, exports and eight charts.
src/data_processing.py	Validates the input; creates observed and optional imputed exports.
src/statistics.py	Welch and rank tests, chi-square, Holm, effect sizes and intervals.
src/visualization.py	Matplotlib/Seaborn charts with labels and denominators.
Stroke_Analysis.ipynb	Guided, executable notebook importing the same functions.
data/stroke.csv	Raw submission dataset.
outputs/tables/	Descriptive, test, proportion and sensitivity CSVs.
build_documents.py	Builds this report, project proposal and viva guide from outputs.
verify_project.py	Checks dataset identity, population counts and statistical computations.

Run the project

1. Extract the ZIP. 2. Open a terminal in Stroke_Capstone_EXCP. 3. Create and activate a Python virtual environment if desired. 4. Run **python -m pip install -r requirements.txt**. 5. Run **python main.py**. 6. Run **python verify_project.py**. 7. Run **python build_documents.py**. The analysis uses the included CSV and does not need a Kaggle account or download after extraction.

Notebook option

Install Jupyter with **python -m pip install jupyterlab**, run **jupyter lab**, then open Stroke_Analysis.ipynb. Run all cells in order from the project folder. It uses the same source modules to avoid separate implementations.

Tested core versions

numpy 2.3.5, pandas 2.2.3, scipy 1.17.0

Data integrity

The SHA-256 digest is recorded in SOURCE.md and outputs/analysis_summary.json. It identifies the exact supplied CSV, including serialization. It does not independently validate clinical provenance.

11. Conclusions, limitations and references

Conclusions supported by this analysis

In this adult educational sample, recorded stroke is associated with older age, higher average glucose and hypertension. The observed mean BMI comparison is inconclusive for a difference, and smoking results are sensitive to incomplete-record handling. All five requested factors are investigated, including results that do not support a simple positive association.

Limitations and possible extensions

The dataset is imbalanced, observational and of limited verified provenance. BMI missingness differs by outcome; smoking contains many Unknown values. There is no temporal information, sampling design or external validation. Covariate-adjusted logistic regression with appropriate handling of missing data could be a future extension, with separate development and validation if prediction becomes a goal. The current work supports an academic data-analysis exercise only.

Capstone assessment coverage

Requirement	Evidence
Problem and dataset (10%)	Sections 1–3; raw CSV and source links.
Modularity (15%)	Three source modules and reusable functions.
NumPy / Pandas (20%)	Validation, filtering, vectorization, grouping and exports.
Visualization (20%)	Eight labeled Matplotlib / Seaborn figures.
Statistical analysis (20%)	SciPy tests, uncertainty, effect sizes and sensitivity checks.
Report (10%)	Methods, evidence and restrained interpretation.
Viva (5%)	Separate Viva_and_Demo_Guide.pdf.

References

[1] fedesoriano. Stroke Prediction Dataset. Kaggle. Source landing page; accessed 25 September 2026.

https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset

[2] Philippe-cz. Stroke-Detection repository. CSV acquisition mirror; accessed 25 September 2026. Source content was downloaded via the GitHub connector.

GitHub mirror: healthcare-dataset-stroke-data.csv

[3] SciPy documentation. Welch t-test: scipy.stats.ttest_ind, equal_var=False.

SciPy: ttest_ind

[4] SciPy documentation. Pearson contingency-table test: scipy.stats.chi2_contingency.

SciPy: chi2_contingency